An Agent-Operated, Market-Anchored Forecasting System

Lessons from the 2026 World Cup Probability Cup

Yifan Luo

Code and data: https://github.com/im-yifanluo/jump-probability-cup

Technical report, October 2026

Abstract

The Jump Trading Probability Cup asked forecasters for probabilities on about ten binary questions per match of the 2026 FIFA World Cup and ranked 4,013 entries by Brier score relative to the field. This report describes the system behind two of those entries and the agent that operated it. The forecasting engine never rates teams itself: it fits a Dixon–Coles scoreline distribution to de-margined bookmaker and prediction-market odds and derives everything else from it—corners, cards, shots and offsides through negative-binomial models regressed on odds-implied scoring rates in about 41,000 club matches, and player props from confirmed lineups—so that answers across 65 question families are mutually consistent. For 38 days a Claude Code agent ran the system in production: it scheduled its own pre-kickoff passes, researched inputs the market misses, submitted and verified both entries, and ran a live A/B test on the second entry under a fixed promotion rule. The stable entry finished 17th overall and the experimental entry 5th in the knockout stage. The relative score decomposes exactly into an *edge* over the crowd's average forecast and a *premium* equal to the dispersion of the field's forecasts. The premium accounts for 90% of the stable entry's points; the edge is positive for both entries and concentrated where the crowd over-prices rare events, but it is not distinguishable from zero over 103 matches. We also report the operating incidents that shaped the system's guardrails and a differential-testing harness that pins the archived code to its tournament behaviour.

Keywords: probabilistic forecasting, Brier score, association football, Dixon–Coles, wisdom of crowds, language-model agents, ML operations

1 Introduction

Forecasting tournaments reward probabilities that are both calibrated and sharp, and the best human forecasters in them combine careful base rates with fast updating on new information (Mellers et al., 2014; Tetlock and Gardner, 2015). In June 2026 Jump Trading sponsored such a tournament for the FIFA World Cup: the Probability Cup, hosted on SportsPredict and run as a recruiting contest (Jump Trading, 2026). Each of the tournament's 104 matches carried about ten yes/no questions—from "Will Argentina win?" to "Will a goal be scored before the first hydration break?"—each answered with a whole-number probability before kickoff and scored by its Brier score relative to the field (SportsPredict, 2026). Bots were allowed, two per person, each ranked as its own entry.

The problem is hard in three specific ways. The questions are *broad*: they span goals, halves, timing windows, corners, cards, shots, offsides, fouls, scorers, player shots on target, substitutions

and advancement, and most have no betting market. The information is *late*: starting lineups appear about an hour before kickoff, the deadline is the kickoff, and kickoff times change from day to day. And the score is *relative*: each forecast is compared with the average of thousands of participants, many of whom read the same public odds.

Our response was to anchor every answer on the betting market, model only what the market does not price, and treat operating the system as an engineering problem delegated to an agent with strict guardrails. The contributions are:

1. A market-anchored pricing engine that answers 65 question families from one odds-fitted scoreline grid plus count and player models, so the answers cannot contradict each other (Section 3).
2. An operating design in which a language-model coding agent runs the production loop (scheduling, research, submission, verification and experimentation) while every submitted number comes out of the statistical pipeline (Section 4).
3. An exact decomposition of relative Brier scoring into an edge over the crowd's average forecast and a premium for the field's dispersion, and an analysis of where the points came from (Sections 5 and 6).
4. A record of the incidents that shaped the operating rules (Section 7) and a differential-testing harness that keeps the archived system reproducible (Section 8).

The stable entry (bot 1) finished #17 of 4,013 overall and #8 in the knockout stage; the experimental entry (bot 2) finished #31 overall and #5 in the knockout stage. On the markets they played, their mean Brier scores were 0.2197 and 0.2174, against 0.2457 for the average participant on bot 1's markets.

2 Background

Scoring. For a market m with outcome $y_m \in \{0, 1\}$ and stage weight w_m (1 in the group stage, 2 in the knockout rounds, 3 in the final), let F_m be the participants who forecast it, $q_{im} \in \{0.01, \ldots, 0.99\}$ their forecasts and $\bar{B}_m = |F_m|^{-1} \sum_{i \in F_m} (q_{im} - y_m)^2$ the field-average Brier score (Brier, 1950). A forecast p earns *relative Brier points*

$$\mathrm{RBP}_m(p) = 100\, w_m (\bar{B}_m - (p - y_m)^2), \tag{1}$$

and the leaderboard ranks the sum over all settled markets. Three consequences shaped the system. First, $\bar{B}_m$ does not depend on p, so the expected score is maximized by minimizing one's own expected Brier score; because the Brier score is strictly proper, the optimal report is one's true probability (Gneiting and Raftery, 2007), and shading toward or away from the crowd can only lose in expectation. Second, a forfeited market scores zero, while any forecast at least as good as the crowd's average scores positive in expectation (Section 5), so coverage has value. Third, late rounds weigh up to three times as much.

Football models and betting markets. Goal counts are commonly modelled as Poisson (Maher, 1982), with the low-score dependence correction of Dixon and Coles (1997) or a bivariate Poisson (Karlis and Ntzoufras, 2003). Betting odds are strong forecasts (Forrest et al., 2005; Spann and Skiera, 2009); machine-learning models struggle to beat closing lines (Hubáček et al., 2019), and for betting, calibration matters more than accuracy (Walsh and Joshi, 2024). Implied probabilities must first be de-margined, proportionally or with Shin's model (Shin, 1993; Štrumbelj, 2014); prediction markets offer a second, independent price (Wolfers and Zitzewitz, 2004). Overdispersed counts such as corners and cards are naturally negative binomial (Cameron and Trivedi, 2013).

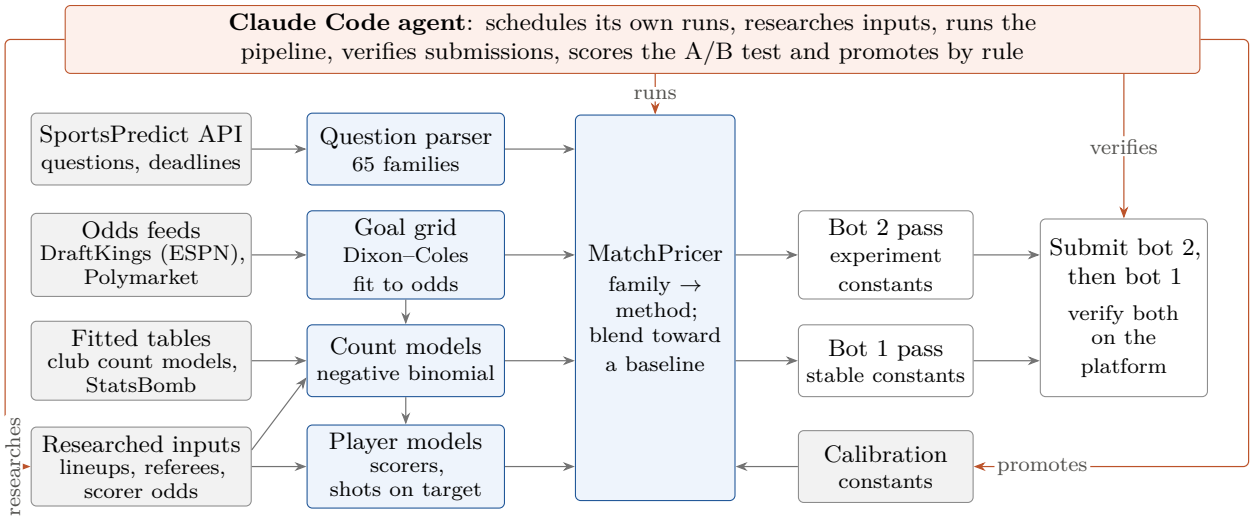


Figure 1. The pipeline. The agent (orange) feeds structured inputs and operates the pipeline; it never edits the models' output. Both bots are priced from identical odds and inputs; bot 2 reprices under its experiment constants and is submitted first, so bot 1's stable values are always the last write.

Crowds. The average of many forecasts is usually better than the average forecaster (Galton, 1907). For squared error this is an identity: the crowd's error equals the average individual error minus the diversity of the forecasts (Krogh and Vedelsby, 1995; Page, 2007). It drives our decomposition in Section 5. Aggregates of independent forecasters often benefit from extremizing (Baron et al., 2014; Satopää et al., 2014); we tested extremizing for our single, already market-anchored forecast and rejected it.

Agents and AI forecasting. Language-model agents that interleave reasoning and tool use (Yao et al., 2023) now operate software environments directly (Yang et al., 2024). Language models have also been evaluated as forecasters, in retrieval-augmented systems (Halawi et al., 2024), live tournaments (Schoenegger and Park, 2023) and dynamic benchmarks (Karger et al., 2025); the Probability Cup itself entered three language models as benchmark bots. Our agent, Claude Code (Anthropic, 2025), has a different role: it does not forecast, but operates a statistical system. That makes the work an instance of the production-ML concerns raised by Sculley et al. (2015), with an online controlled experiment (Kohavi et al., 2020) on the second entry, and with differential testing (McKeeman, 1998) and characterization tests (Feathers, 2004) to archive it.

3 The forecasting engine

Figure 1 shows the pipeline. Its organizing rule is a pricing hierarchy: (1) a market price for this exact fixture wins where one exists; (2) otherwise, the output of models fitted to that price; (3) otherwise, conditional tables measured on international-tournament event data (StatsBomb); (4) only then a base rate, which is always flagged.

3.1 Anchoring on the market

For each fixture we take the 1X2 (home, draw, away) prices and the over/under line from the ESPN feed of DraftKings odds and remove the margin proportionally, $\pi_k = (1/o_k)/\sum_j (1/o_j)$ (Štrumbelj, 2014). Where Polymarket lists the match, its de-margined 1X2 is averaged in, and disagreements

above 10 points are flagged as likely stale. The total expected goals T solves $\Pr(N \geq L + \frac{1}{2}) = \pi_{\text{over}}$ for $N \sim \text{Poisson}(T)$ and line L. A dated manual override exists for fixtures without a feed; it expires loudly on any other date.

3.2 The goal grid

Home and away goals (X, Y) follow independent Poisson laws with the Dixon–Coles correction (Dixon and Coles, 1997),

$$\Pr(X = x, Y = y) \propto \tau(x, y)\, \frac{\lambda_H^x e^{-\lambda_H}}{x!}\, \frac{\lambda_A^y e^{-\lambda_A}}{y!}, \tag{2}$$

with $\tau(0,0) = 1 - \lambda_H \lambda_A \rho$, $\tau(1,0) = 1 + \lambda_A \rho$, $\tau(0,1) = 1 + \lambda_H \rho$, $\tau(1,1) = 1 - \rho$ and $\tau = 1$ elsewhere, on a grid truncated at 12 goals and renormalized. The parameters are not estimated from team histories. With $\lambda_A = T - \lambda_H$, we solve for (λ_H, ρ) such that the grid reproduces the market's home-win and draw probabilities exactly, with ρ clamped to keep every cell non-negative. Every goal question is then a sum over cells of one grid: results, totals, both teams scoring, margins, clean sheets and combinations ("both teams score and 3+ goals" is a joint sum, never a product of marginals). Half-based questions split each rate into halves with the second-half share $h = 0.60$ measured on international tournaments (club football gives 0.54). The first goal is a Poisson race, $\Pr(\text{team scores first}) = \lambda_{\text{team}}/(\lambda_H + \lambda_A) \cdot (1 - \Pr(0,0))$; a large-sample check on 219 tournament matches confirmed that goals are exchangeable in time (observed 0.886, model 0.878). Advancement in a knockout tie adds extra time and penalties explicitly:

$$\Pr(A \text{ advances}) = \Pr(X_A > X_B) + \Pr(X_A = X_B)\big[\Pr(E_A > E_B) + \Pr(E_A = E_B)\, s\big], \tag{3}$$

where E_A, E_B are Poisson extra-time goals at a third of each side's 90-minute rate and the shoot-out share $s = \frac{1}{2} + \kappa(\sigma_A - \frac{1}{2})$ shrinks the regulation strength share $\sigma_A = \Pr(X_A > X_B)/(\Pr(X_A > X_B) + \Pr(X_A < X_B))$ toward a coin flip ($\kappa = 0.25$). The two sides' probabilities sum to one exactly, and the shoot-out market uses the same extra-time Poissons.

3.3 Count models

Corners, cards, shots on target (SOT) and fouls use negative-binomial (NB2) models fitted on club football from football-data.co.uk (Football-Data.co.uk): the 40,596 matches before the 2025/26 season (out of 115,987 in 19 leagues) that have both match statistics and closing 1X2 and over/under odds. The odds give each match the same odds-implied rates $(\lambda_{\text{team}}, \lambda_{\text{opp}})$ that the live system computes:

$$\log \mu_k = \beta_{0k} + \beta_{1k} \log \lambda_{\text{team}} + \beta_{2k} \log \lambda_{\text{opp}} + \beta_{3k}\, \text{home}, \qquad \text{Var} = \mu_k(1 + \alpha_k \mu_k). \tag{4}$$

The coefficients come from a Poisson GLM and α_k from the moment estimator of Cameron and Trivedi (2013) (Table 1). Because the covariates are market-implied rather than team identities, the fit transfers to any fixture with odds. At the World Cup the venue is neutral (home $= \frac{1}{2}$) and each mean is multiplied by tournament corrections (below) and, for cards, by the referee's strictness, (referee cards per game)/3.46, capped at 1. Totals are exact convolutions of the two team distributions (a sum of unequal-mean NBs is not NB), and "more X than" comparisons enumerate both, with ties resolving to No. Offsides have no club coverage and depend more on defensive lines than on attacking strength: their mean is the tournament team mean times $(\lambda_{\text{team}}/1.30)^{0.18}$, an elasticity measured on event data. Timing-window questions ("a goal before the first hydration break") use the share of each event type falling in that window, measured from event timestamps, applied to the match mean.

Table 1. Count models fitted on club matches (Equation (4)). Shots on target scale almost proportionally with a team's own odds-implied scoring rate; cards and fouls fall as a team gets stronger.

Statistic	β_0	β_1 (own rate)	β_2 (opponent)	β_3 (home)	α
Corners	1.468	0.451	−0.156	0.057	0.078
Cards	0.895	−0.379	−0.119	−0.060	0.018
Shots on target	1.180	0.797	0.054	0.038	0.031
Fouls	2.668	−0.278	−0.240	−0.007	0.025

3.4 Players

A player's anytime-scorer probability is $1 - \exp(-g\,\lambda_{\text{team}}\,m/90)$, where g is the player's share of team goals, shrunk toward 0.12 with a prior weight of six team goals, and m is the player's expected minutes. Minutes come from the confirmed starting XI, which the pipeline pulls from ESPN as soon as it is posted; a benched player's minutes drop automatically. Shots on target use a per-match rate r: a club rate times 0.839 (tournament play is less open), shrunk toward 0.72 with the weight of one appearance, then multiplied by the match tempo $\min(\mu^{\text{SOT}}_{\text{team}}/4.11,\, 1.6)$ and by $m/90$. For the threshold of one SOT and a genuine shooter ($r \geq 0.5$) the tempo swing is halved, because a starter shoots regardless of the opponent; this lowered held-out Brier from 0.275 to 0.255 on 234 shooter-matches. A leave-one-tournament-out check of the SOT model on 4,989 player-matches gave a Brier score of 0.174, against 0.212 for the previous model and 0.250 for a coin flip.

Scorer markets are the one place where a researched number replaces the model: a de-margined anytime-scorer price for this exact fixture, $p = (1/o)/1.10$, tagged with the fixture name and expiring loudly if reused for another match. For a star confirmed on the bench, the agent prices the cameo explicitly, as the probability of coming on times the probability of scoring in the expected minutes. Score-or-assist markets are kept consistent with these anchors by a sweep: each must sit at or above the anytime price plus an assist uplift.

3.5 Tuned constants and coverage

Every tuned constant lives in one module, which the models read at call time; that is what lets bot 2 reprice under different constants (Section 4.3). Table 2 lists the corrections that ran in the final. They are multiplicative and *level-specific*: a per-team scale also moves totals, but a total-only scale never touches per-team or comparison markets, which are scale-invariant. An early uniform "low-event" multiplier on every count statistic dropped bot 1 from about 50th to 94th within days; per-statistic corrections, validated out of sample, replaced it. Families whose models showed weak out-of-sample skill are shrunk toward the same machinery run at tournament-average rates, $p = p_{\text{base}} + w\,(p_{\text{model}} - p_{\text{base}})$, with weights from the club backtest (0.40 for card thresholds, 0.67 for "more goals in the second half").

Questions arrive as free text. About a hundred small rules map each to a structured specification (family, team, player, statistic, threshold, period); all 1,153 questions on the 101 archived boards parse into 65 families (Table 10). Anything unrecognized is logged and never priced by guesswork. When the knockout boards re-phrased every question with "in regulation (90 minutes + stoppage time)", a normalizer let the group-stage rules serve them, and 19 new families were built and tested during the knockouts, each on the day it appeared.

Table 2. The tournament corrections that ran in the final (all in `models/calibration.py`).

Constant	Value	Scope	Evidence
Tournament intensity	cards 0.78, SOT 1.00, corners 1.05, offsides and fouls 0.90	per-team means	Re-fitted per statistic on 383 settled markets with chronological out-of-sample validation, replacing a uniform 0.90
Total SOT scale	1.50	totals only	The platform counts about 1.5× the event-data total (implied 1.49, 95% CI 1.24–1.78); bot-2 probe won 4 of 5
Total cards scale	0.85	totals only	Cards over-predicted by 0.05 on 724 markets; bot-2 probe won 8 of 13
Total offsides scale	1.10	totals only	Consistent under-prediction of totals
Favourite SOT boost	0.06	favourite only	Club elasticity too flat for mismatches; probe won 4 of 4 divergent markets
Timing scale	0.85	timing windows	Realized 0.214 vs 0.432 priced ($n = 28$); probe 5 of 6; walk-forward −1.83 Brier points
Referee factor	≤ 1	cards	Strict referees over-predicted cards; only leniency is priced
Shoot-out tilt κ	0.25	advancement	Shoot-outs are close to coin flips

4 Operating the system with an agent

Table 3 divides the work. The statistical pipeline produces every probability. The agent, a Claude Code session working from three checklist playbooks, does everything around it, and the operator (the author) sets objectives and rules.

4.1 The daily cycle

Kickoff times moved every day, so fixed-time runs kept missing the short window between the lineup release (about 60 minutes before kickoff) and the deadline (kickoff). The agent therefore scheduled its own day (Figure 2). A morning run scored the previous day's markets, scored the A/B

Table 3. Division of labour.

Role	Responsibility
Pipeline	Every probability: odds-anchored goal grid, count and player models, one pricer per match.
Agent	Inputs the market misses (confirmed lineups, referee strictness, de-margined scorer prices), written to match-scoped files; running the pipeline on schedule; verifying submissions; running and scoring the A/B bench; reports and long-term memory notes.
Operator	Objectives and rules: standing authorization for bot-1 submissions (June 12) and for promoting validated probes (July 2); quality over coverage; the final-day objectives.

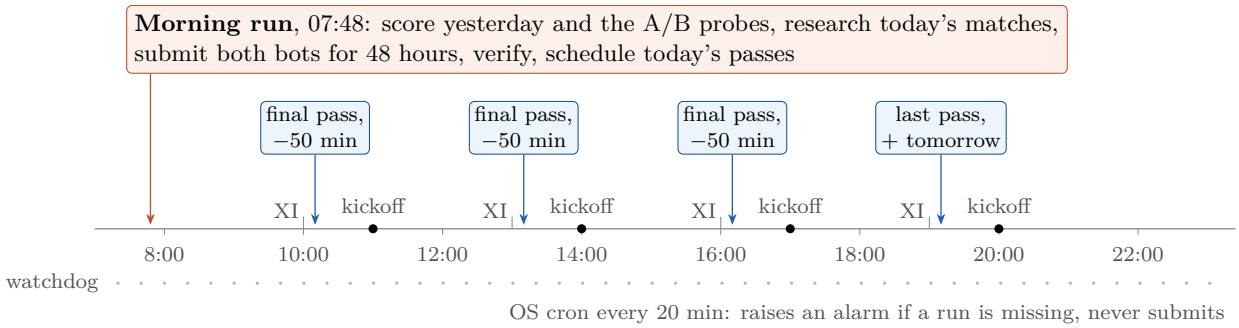


Figure 2. An illustrative match day with four kickoff waves (local time). The morning run schedules one final pass 50 minutes before each wave, just after the confirmed lineups (XI) appear; the last pass also prices the next day. A durable operating-system job checks every 20 minutes that each scheduled run happened and raises an alarm if not.

probes, researched the day's matches, submitted both bots for the next 48 hours, verified them and printed one-shot schedule entries for 50 minutes before each kickoff wave. Each pass then pulled the confirmed lineups, the final odds and the referee and resubmitted; the last pass of the day also priced the next day. A late pass costs freshness, never correctness: the pipeline refuses to write to a market whose kickoff has passed.

4.2 Guardrails

- **Inputs, not numbers.** Research enters only through structured files, each entry scoped to one match; an input reused for another fixture expires loudly.
- **Contamination-safe order.** Both bots are priced from the same odds; bot 2 is submitted first and bot 1 last, so the stable values are always the final write.
- **Verify, don't trust the summary.** After every pass the agent queries both entries on the platform and compares their open-prediction counts.
- **Quality over coverage.** No mechanical fallback ever submits. A skipped refresh costs only freshness, because the previous submission stands; a careless one can cost points. A missed run raises an alarm and a researched run is triggered.
- **Memory outside the context window.** Sessions restart, so decisions, root causes and constants were kept in 21 short notes indexed in a file that loads at the start of every session, alongside the playbooks, the strategy document and the logs.

4.3 The A/B bench

Bot 2 tested one family of changes at a time against bot 1's configuration. A context manager swaps any of eight calibration constants for one pricing pass and always restores them, so both bots see identical odds and inputs and the strategy is the only difference. A probe was promoted to bot 1 after it won about 8–10 of its own settled markets in a consistent direction, counting only markets settled *after* bot 1's last change to that constant (a tally that ignored this once credited a probe with 10 wins out of 13 when its clean sample was empty). Live drift flags were treated as hypotheses for large-sample tests, never as fixes: three of the four raised in the knockouts (first goal, both teams scoring, knockout tempo) were cleared on event or club data. Table 4 lists the decisions.

Table 4. Probes run on bot 2 and their outcomes. "Won k of n" counts settled markets where the two bots differed.

Probe	Dates	Evidence	Outcome
Stronger count multipliers	Jun 26–29	Trailed bot 1 by about 0.015 Brier on 33 differing markets	Rejected
Total SOT scale 1.40	Jun 29	Same-day natural experiment; total-SOT leave-out Brier 0.245 → 0.142	Promoted (totals only)
Total SOT scale 1.60	Jun 29–Jul 2	Won 4 of 5 against 1.40; implied scale 1.49	Bot 1 set to 1.50
Total cards 0.85, offsides 1.10, favourite SOT 0.06	Jul 1–2	Won 8 of 13 card settles and 4 of 4 favourite-SOT divergents; calibration corroborated	Promoted
Total shots 1.10–1.20	Jul 1–7	Net loss	Retired
Timing scale 0.85	Jul 3–10	Won 5 of 6; calibration gap +0.146; walk-forward backtest	Promoted
Next notches: cards 0.80, timing 0.70, total SOT 1.60, favourite 0.10, offsides 1.20	Jul 3–19	Timing 0.70 lost 5 of 5; cards 0.80 net loss ($n = 11$); others flat or negative	Retired

4.4 Activity

Over 38 days the agent ran the production pipeline 254 times (201 submitting runs and 53 dry runs) and logged 17,005 submission actions: 2,158 creates, 3,986 updates, 10,711 checks that left a prediction unchanged and 149 errors (Figure 3). It wrote 110 run reports and an experiment log with every verdict.

5 Evaluation method

Data. After the event the platform's results endpoint returned, for every settled market, our probability, the crowd's average probability $\bar{q}_m$ (rounded to a whole percent), the outcome and the stage-weighted points. The export covers 1,183 markets in 103 matches for bot 1 and 1,035 in 88 matches for bot 2. Stage weights come from the question wording (knockout questions carry the regulation qualifier) and the final's match identifier.

Decomposition. For squared error the field-average score splits into the score of the average forecast and the spread of the forecasts around it (Krogh and Vedelsby, 1995; Page, 2007): $\bar{B}_m = (\bar{q}_m - y_m)^2 + V_m$ with $V_m = |F_m|^{-1} \sum_{i \in F_m} (q_{im} - \bar{q}_m)^2$. Substituting into Equation (1),

$$\mathrm{RBP}_m(p) = \underbrace{100\, w_m \left[(\bar{q}_m - y_m)^2 - (p - y_m)^2 \right]}_{\text{edge } E_m} + \underbrace{100\, w_m V_m}_{\text{premium } P_m}. \tag{5}$$

The premium does not depend on our forecast; every participant who forecasts market m receives it on top of their own edge. Every other term in Equation (5) is in the export, so V_m can be recovered market by market, assuming the platform computes the crowd's average and the field-average score over the same participants. It must be non-negative, which checks the stage weights: with the

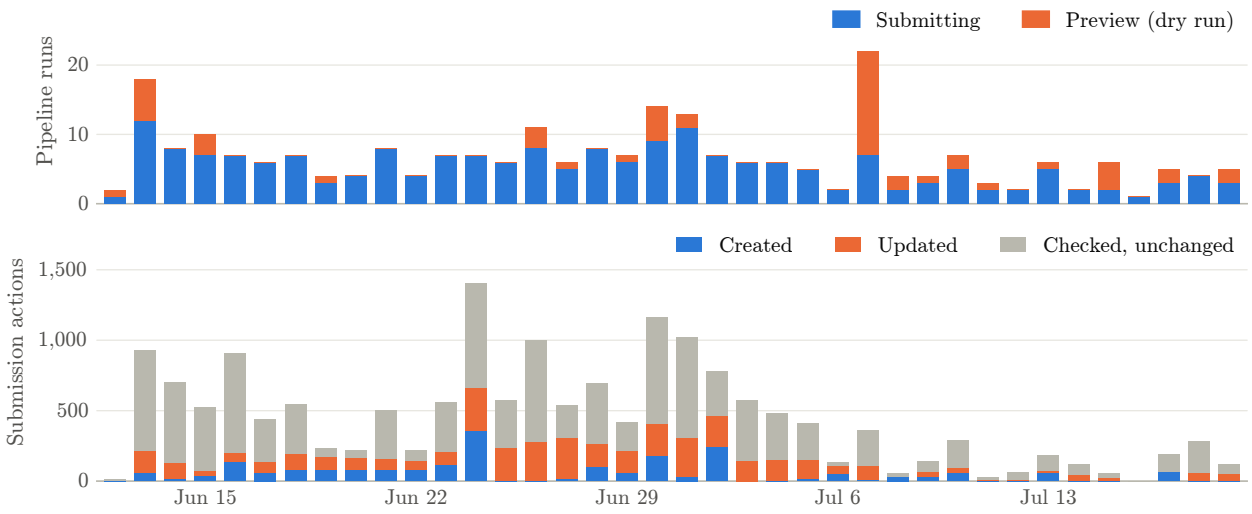


Figure 3. Daily operation, June 12 to July 19. Top: production-pipeline runs. Bottom: submission actions across both entries; most were idempotent checks of predictions already in place.

weights above, the recovered V_m is positive on all 2,218 markets. Rounding $\bar{q}_m$ to a whole percent perturbs each V_m by less than 0.01. The average recovered V_m is 0.0251: a typical participant's forecast sits about 16 points from the crowd's average. Note that the premium is not available to any single forecaster for free: the crowd's average was not visible before kickoff.

Uncertainty. Markets within a match settle together (one lopsided match decides a dozen of them), so we resample whole matches (Cameron et al., 2008; Efron, 1979), with 4,000 resamples, and report 95% percentile intervals.

Calibration. We report reliability diagrams, the expected calibration error over ten equal-width bins (Guo et al., 2017) and the corresponding binned Murphy decomposition of the Brier score into reliability, resolution and uncertainty (DeGroot and Fienberg, 1983; Murphy, 1973).

Offline validation. Before the tournament, a walk-forward backtest rebuilt the pricer for every club match with closing odds in nine test seasons (2017/18 to 2025/26), with count models fitted only on earlier seasons, and scored 19 synthetic question families against what happened: 782,157 forecasts in all (Table 9).

6 Results

6.1 Where the points came from

Table 5 applies Equation (5) to both entries. Of bot 1's 4,732 points, 4,275 (90%) are premium and +457 are edge over the crowd's average forecast. Bot 2's edge is larger (+631), but neither interval excludes zero. The mean Brier scores tell the same story: bot 1's forecasts (0.2197) are slightly better than the crowd's average forecast on the same markets (0.2206), and both are far better than the average participant (0.2457), who scores barely better than a constant 50% (0.25). Bot 1 scored above the field average on 69.1% of its markets but was closer to the outcome than the crowd's average forecast on only 57.0%.

Table 5. Points decomposed into premium (the field's dispersion) and edge over the crowd's average forecast (Equation (5)), with 95% intervals from resampling matches. Bot 1 played 1,183 markets in 103 matches; bot 2 played 1,035 in 88. The mean Brier scores are ours, the crowd's average forecast's, and the average participant's.

				Mean Brier score		
Entry	Points	Premium	Edge	Ours	Crowd avg.	Field
Bot 1 (stable)	4,732	4,275	+457	0.2197	0.2206	0.2457
	[3,299, 6,342]	[3,672, 4,908]	[−751, 1,754]			
Bot 2 (A/B bench)	4,451	3,820	+631	0.2174	0.2194	0.2437
	[3,072, 6,020]	[3,259, 4,475]	[−527, 1,846]			

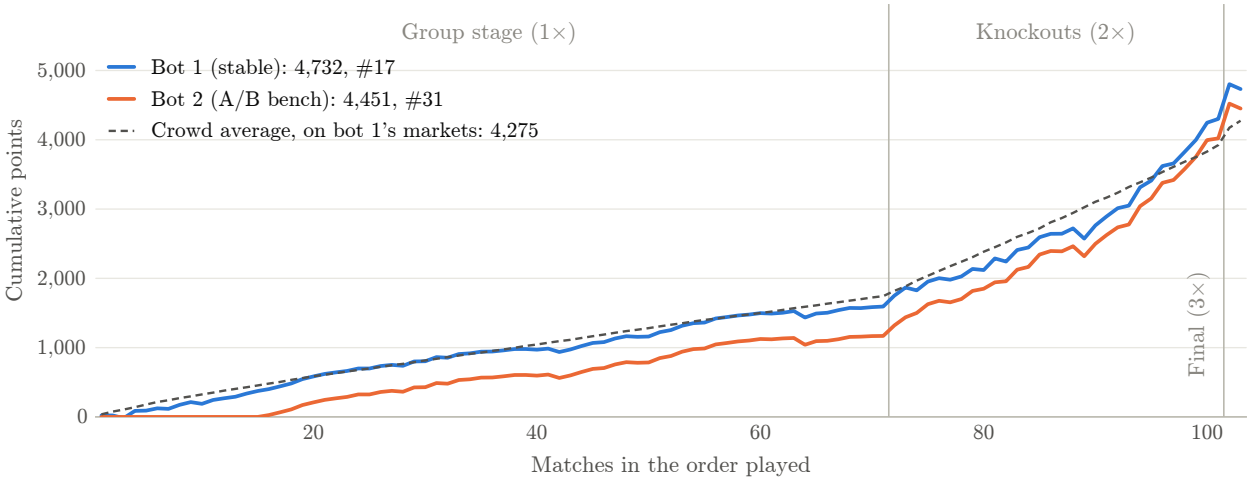


Figure 4. Cumulative points over the 103 matches. The dashed line is the premium alone on bot 1's markets: what submitting the crowd's average would have earned. Bot 2 earned nothing on its first 15 matches (Section 7).

Two consequences follow. Among entrants who forecast the same markets the premium is common, so their ranking depends only on their own Brier scores. Across entrants who forecast different markets the premium rewards participation: about 3.61 points per market for bot 1, provided the forecast is no worse than the crowd's average. A hypothetical entrant who submitted the crowd's average on every market bot 1 played would have scored 4,275, below both bots (Figure 4); where that score would have ranked is unknown, because the platform does not publish the distribution of totals. Two-thirds of bot 1's points (66%) came after the group stage, and the final alone, at triple weight, was worth 501.6 and moved it from 31st to 17th.

6.2 Where the edge came from

The crowd's average forecast is biased upward: across bot 1's markets it averaged 47.1% against a realized rate of 42.2%, while bot 1 averaged 42.2%. Table 6 and Figure 5a break this down. The crowd over-priced events most on cards (+13.3 points), player shots on target (+12.4), penalties (+10.9) and goal scorers (+10.5). Bot 1 was much less biased on all of these except penalties, where both followed similar base rates. Its edge per market was largest on timing windows (+2.9), cards (+1.7) and player shots on target (+1.1); on goal scorers the smaller bias earned little (+0.3

Table 6. Bot 1 by market type: points, edge over the crowd's average forecast, and bias (mean forecast minus realized rate, in percentage points) for bot 1 and for the crowd's average.

					Bias (points)	
Market type	Markets	Points	Per market	Edge	Bot 1	Crowd
Goals, results & halftime	379	+1,462	+3.9	+279	−2.1	−0.8
Player shots on target	116	+662	+5.7	+122	+0.4	+12.4
Goal scorers	121	+627	+5.2	+40	+4.7	+10.5
Cards	90	+479	+5.3	+155	+6.6	+13.3
Corners	85	+362	+4.3	+70	+1.0	+6.4
Team & total shots on target	158	+325	+2.1	−215	−5.7	+3.5
Timing windows	29	+243	+8.4	+85	+5.0	+7.3
Penalties	55	+184	+3.4	+33	+11.3	+10.9
Offsides	74	+152	+2.1	−82	−3.5	+3.6
Other	7	+106	+15.2	+62	+24.3	+26.1
Fouls	56	+80	+1.4	−43	−5.6	−0.8
Total shots	13	+49	+3.8	−48	+0.8	−8.2
All markets	1,183	+4,732	+4.0	+457	0.0	+4.9

Table 7. Bot 1 by disagreement with the crowd, d = bot 1's probability minus the crowd's average, in points. Mean forecasts and the realized rate are in percent; points, premium and edge are per market.

Disagreement	Markets	Bot 1	Crowd	Realized	Points	Premium	Edge
$d < -15$	129	20	42	29	+6.5	+5.4	+1.2
$-15 \le d < -5$	377	36	46	37	+4.7	+3.5	+1.1
$-5 \le d < 5$	558	47	48	47	+3.5	+3.3	+0.2
$5 \le d < 15$	101	61	53	52	+2.0	+3.3	−1.3
$d \ge 15$	18	74	55	44	−2.0	+4.8	−6.8

per market). On goals and results the crowd is nearly unbiased, because both follow the same betting market; the edge there is modest per market (+0.7) but, over 379 markets, the largest in total. The losses are informative too: bot 1 under-predicted team and total shots on target by 5.7 points on average (−215 edge), the leak that the total-SOT scale of Table 2 was introduced to close mid-tournament, and it also under-predicted offsides and fouls.

Table 7 and Figure 5b group bot 1's markets by its disagreement with the crowd, $d = 100(p - \bar{q})$. Points per market fall steadily from +6.5 when bot 1 sat more than 15 points below the crowd to −2.0 when it sat 15 or more above. The edge column shows why: pricing below the crowd beat the crowd's average forecast, pricing above it lost, consistent with the crowd's upward bias. The premium is roughly constant across bins, so even forecasts within five points of the crowd earned +3.5 per market.

6.3 Calibration

In the club backtest the pricer tracks the diagonal within two percentage points in every bin from 5% to 95% (Figure 6a), and it beat a constant 50% forecast in all 19 families and the historical

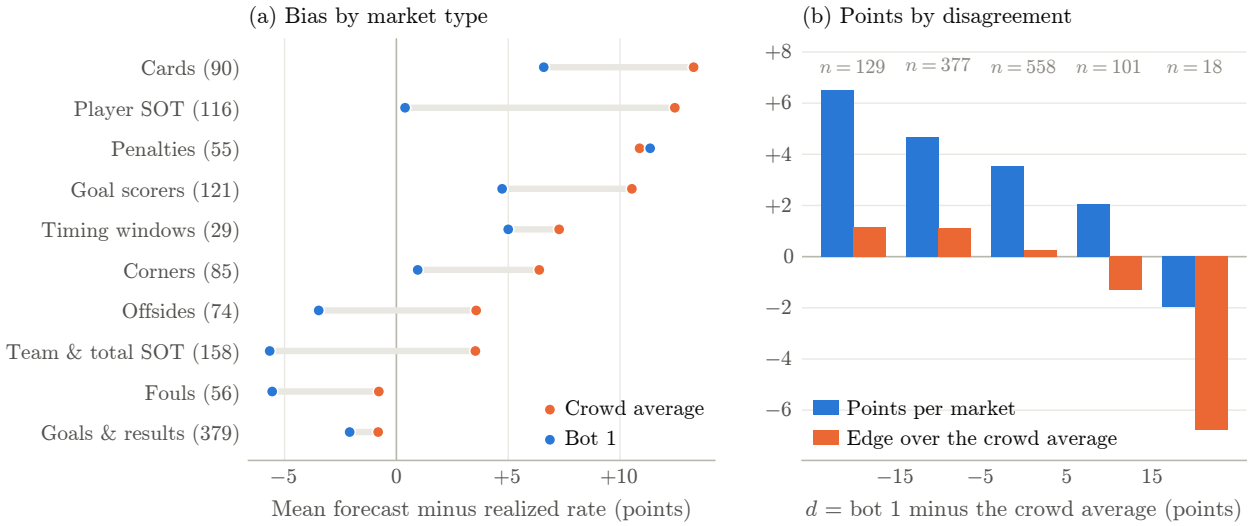


Figure 5. (a) Bias by market type (types with at least 25 markets; SOT = shots on target): the crowd's average over-prices most event types, bot 1 much less. (b) Points and edge per market by how far bot 1 sat from the crowd's average.

base rate in 17 (Table 9). In the tournament (Figure 6b) bot 1's expected calibration error was 0.030 against 0.057 for the crowd's average. In the Murphy decomposition the crowd's average is less reliable (0.0051 against 0.0018) but slightly more resolved (0.0283 against 0.0239): it separates outcomes a little better while being shifted upward. A share of the crowd's information is therefore not in the bots' inputs, which is an argument for the researched inputs the agent supplied, and for more of them.

6.4 The A/B bench

On the 1,035 markets both bots played, their forecasts differed on 112 markets in 36 matches. Bot 2 was closer to the outcome on 59 of them and bot 1 on 53, a net +93.7 points for bot 2 (95% interval −215 to +421); over all common markets the mean Brier scores were 0.2176 and 0.2174. The 280.8-point gap between the bots' totals is entirely the 148 markets in the 15 matches bot 2 missed (374.5 points), less bot 2's gain on the divergent markets. The bench's measurable contribution is therefore small. Its value was procedural: every change reached the ranked entry through it, and it caught harmful ideas, such as the stronger count multipliers, before they reached bot 1.

6.5 Reproducing the final

The archived code reprices the final's 20 markets offline from odds recovered from four goal markets; 19 of the 20 forecasts equal what bot 1 submitted to the whole percent, and the twentieth differs by one point (`probcup demo`). This is direct evidence that the archive is the system that ran.

7 Incidents and lessons

Table 8 lists the failures that changed the system. Most were operational rather than statistical, and each left a rule behind.

Three lessons generalize. *Verify outcomes, not reports*: the costliest failure was silent, because the submission summary was computed from the wrong state; only querying the platform revealed

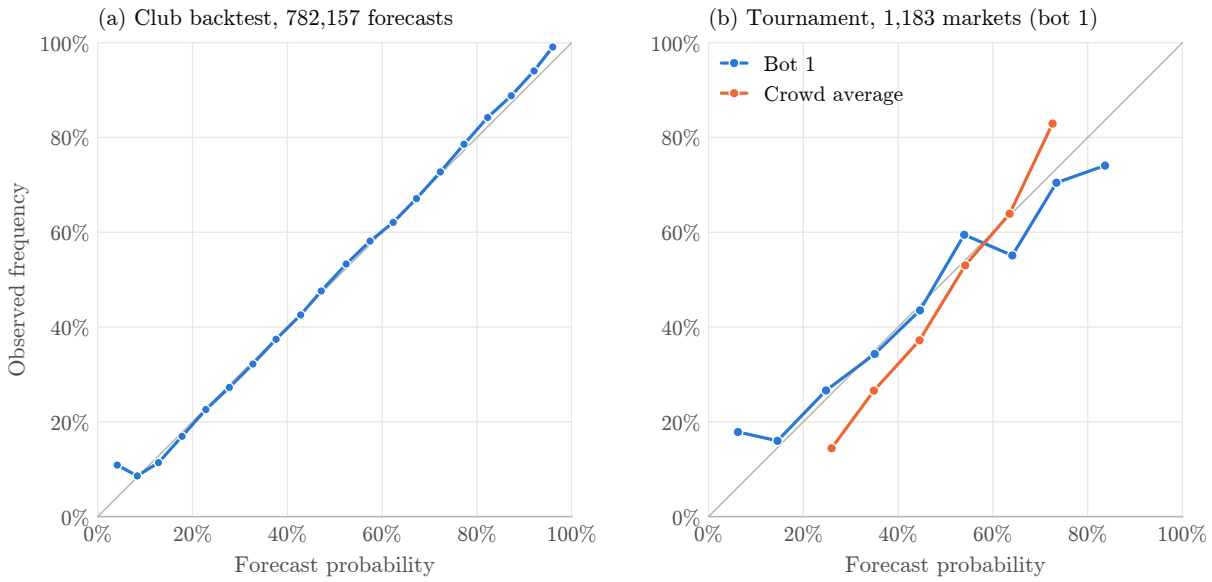


Figure 6. Reliability. (a) Walk-forward club backtest in 5-point bins. (b) The tournament in 10-point bins with at least 20 markets: bot 1 and the crowd's average forecast on the same markets.

Table 8. Incidents and the rules they left.

Date	What happened	Rule that stayed
Jun 13	The first scheduled run failed: commands register only at session start.	Scheduled prompts are self-contained ("read the playbook and execute it").
Jun 16	Bot 2 had been empty since launch while every pass reported it unchanged: the dual submission read bot 1's state. About 15 matches were lost.	Each bot is submitted its own values; every pass verifies both on the platform.
Jun 24	A uniform low-event multiplier on every count statistic dropped bot 1 from about #50 to #94.	Per-statistic corrections, validated out of sample; never a uniform global dial.
Jun 26	A stale "benched" override priced a starter at 5% to register a shot on target.	Lineups are pulled automatically; inputs are scoped to one match; a staleness guard.
Jun 28	Knockout boards re-phrased every question and the parser rejected them.	Phrasing normalization; new families built and tested the same day; a backlog scanner.
Jul 7	A probe looked like a 10-of-13 winner, but the wins predated bot 1's last change.	Only settlements after the last change count toward promotion.
Jul 11	A restart wiped the session's schedule; a morning run and a final pass never fired.	Two layers: session schedules run the researched passes; a durable job only raises alarms.
Jul 15	A stale score-or-assist share on a prolific forward cost 94 points, the worst single market.	The score-or-assist consistency sweep on every board.

it. *Fail loudly rather than fall back*: when runs were missed, the tempting fix was an automatic submission, but under relative scoring a careless forecast can lose points while a skipped refresh loses only freshness, so the durable layer raises alarms and never submits. *Change the ranked entry only on large samples*: live samples were small and correlated within matches, and three of four drift flags, as well as a promotion tally, turned out to be noise once tested properly.

8 Reproducibility

The repository is the archived system, restructured into an installable package with one command-line entry point. Before anything moved, the code that ran the final (tag `tournament-final`) was run against a fake outside world (stateful fakes of the SportsPredict, ESPN, Polymarket and API-Football endpoints and a frozen clock) and every observable effect was recorded: 1,153 real markets priced for both bots, and thirteen command scenarios down to each HTTP request, submitted value, printed line and file written. The restructured code reproduces all of it exactly, and the suite runs offline in about 30 seconds in continuous integration. The ten research studies and the full club backtest also produce identical output before and after.

Every tournament result in this report (numbers, tables and figures) is generated from the committed results export by `paper/build_assets.py`, and `make paper` rebuilds the report from scratch; figures quoted from the research studies are in their committed outputs. The fitted models and researched inputs as they stood for the final are committed; the raw event and club data are downloaded by `probcup download-data`.

9 Limitations

The evidence comes from one tournament of 103 matches, and the edge over the crowd is not statistically distinguishable from zero. We cannot compute counterfactual ranks, because the distribution of leaderboard totals is not public, and the crowd's average is rounded to a whole percent. The crowd data available *during* the tournament were copied by hand from 16 matches; the full export only became available afterwards, so mid-tournament decisions rested on far less evidence than this report. The quality of the agent's research inputs is not measured separately from the models'. Finally, the archive deliberately keeps the defects that ran live; the most consequential is that "Will a substitute score or assist?" was parsed as a player named "a substitute" and priced at a generic prior, including in the final.

10 Conclusion

Relative Brier scoring rewards two things: forecasting every market at least as well as the crowd's average, which earns the field's dispersion, and beating that average where the crowd is biased. A market-anchored engine delivers the first by construction; careful models of the event types the crowd over-prices deliver the second, though over one tournament that edge is small and uncertain. The larger lesson concerns operation. An agent can run a forecasting system in production for weeks, but only inside rules that make its failures loud: verified submissions, match-scoped inputs, alarms instead of fallbacks, and a separate entry for experiments.

Acknowledgements and data

Event data are from StatsBomb Open Data (StatsBomb); club results and closing odds from football-data.co.uk (Football-Data.co.uk); live odds from ESPN (DraftKings) and Polymarket; player season rates from API-Football. This work is not affiliated with Jump Trading, SportsPredict or FIFA.

References

Anthropic. Claude Code. https://docs.anthropic.com/en/docs/claude-code, 2025. Agentic coding tool.

Jonathan Baron, Barbara A. Mellers, Philip E. Tetlock, Eric Stone, and Lyle H. Ungar. Two reasons to make aggregated probability forecasts more extreme. *Decision Analysis*, 11(2):133–145, 2014. doi: 10.1287/deca.2014.0293.

Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1): 1–3, 1950. doi: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.

A. Colin Cameron and Pravin K. Trivedi. *Regression Analysis of Count Data*. Cambridge University Press, second edition, 2013. doi: 10.1017/CBO9781139013567.

A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427, 2008. doi: 10.1162/rest.90.3.414.

Morris H. DeGroot and Stephen E. Fienberg. The comparison and evaluation of forecasters. *Journal of the Royal Statistical Society: Series D (The Statistician)*, 32(1–2):12–22, 1983. doi: 10.2307/2987588.

Mark J. Dixon and Stuart G. Coles. Modelling association football scores and inefficiencies in the football betting market. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 46(2):265–280, 1997. doi: 10.1111/1467-9876.00065.

Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. doi: 10.1214/aos/1176344552.

Michael C. Feathers. *Working Effectively with Legacy Code*. Prentice Hall, 2004.

Football-Data.co.uk. Historical football results and betting odds data. https://www.football-data.co.uk/.

David Forrest, John Goddard, and Robert Simmons. Odds-setters as forecasters: The case of English football. *International Journal of Forecasting*, 21(3):551–564, 2005. doi: 10.1016/j.ijforecast.2005.03.003.

Francis Galton. Vox populi. *Nature*, 75:450–451, 1907. doi: 10.1038/075450a0.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017.

Danny Halawi, Fred Zhang, Chen Yueh-Han, and Jacob Steinhardt. Approaching human-level forecasting with language models. In *Advances in Neural Information Processing Systems 37*, 2024. arXiv:2402.18563.

Ondřej Hubáček, Gustav Šourek, and Filip Železný. Exploiting sports-betting market using machine learning. *International Journal of Forecasting*, 35(2):783–796, 2019. doi: 10.1016/j.ijforecast.2019.01.001.

Jump Trading. Introducing the Jump Trading Probability Cup. https://www.jumptrading.com/signals/introducing-the-jump-trading-probability-cup, 2026.

Ezra Karger, Houtan Bastani, Chen Yueh-Han, Zachary Jacobs, Danny Halawi, Fred Zhang, and Philip E. Tetlock. ForecastBench: A dynamic benchmark of AI forecasting capabilities. In *International Conference on Learning Representations*, 2025. arXiv:2409.19839.

Dimitris Karlis and Ioannis Ntzoufras. Analysis of sports data by using bivariate Poisson models. *Journal of the Royal Statistical Society: Series D (The Statistician)*, 52(3):381–393, 2003. doi: 10.1111/1467-9884.00366.

Ron Kohavi, Diane Tang, and Ya Xu. *Trustworthy Online Controlled Experiments: A Practical Guide to A/B Testing*. Cambridge University Press, 2020. doi: 10.1017/9781108653985.

Anders Krogh and Jesper Vedelsby. Neural network ensembles, cross validation, and active learning. In *Advances in Neural Information Processing Systems 7*, pages 231–238. MIT Press, 1995.

Michael J. Maher. Modelling association football scores. *Statistica Neerlandica*, 36(3):109–118, 1982. doi: 10.1111/j.1467-9574.1982.tb00782.x.

William M. McKeeman. Differential testing for software. *Digital Technical Journal*, 10(1):100–107, 1998.

Barbara Mellers, Lyle Ungar, Jonathan Baron, Jaime Ramos, Burcu Gurcay, Katrina Fincher, Sydney E. Scott, Don Moore, Pavel Atanasov, Samuel A. Swift, Terry Murray, Eric Stone, and Philip E. Tetlock. Psychological strategies for winning a geopolitical forecasting tournament. *Psychological Science*, 25(5): 1106–1115, 2014. doi: 10.1177/0956797614524255.

Allan H. Murphy. A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4): 595–600, 1973. doi: 10.1175/1520-0450(1973)012<0595:ANVPOT>2.0.CO;2.

Scott E. Page. *The Difference: How the Power of Diversity Creates Better Groups, Firms, Schools, and Societies*. Princeton University Press, 2007.

Ville A. Satopää, Jonathan Baron, Dean P. Foster, Barbara A. Mellers, Philip E. Tetlock, and Lyle H. Ungar. Combining multiple probability predictions using a simple logit model. *International Journal of Forecasting*, 30(2):344–356, 2014. doi: 10.1016/j.ijforecast.2013.09.009.

Philipp Schoenegger and Peter S. Park. Large language model prediction capabilities: Evidence from a real-world forecasting tournament, 2023. arXiv:2310.13014.

D. Sculley, Gary Holt, Daniel Golovin, Eugene Davydov, Todd Phillips, Dietmar Ebner, Vinay Chaudhary, Michael Young, Jean-François Crespo, and Dan Dennison. Hidden technical debt in machine learning systems. In *Advances in Neural Information Processing Systems 28*, pages 2503–2511, 2015.

Hyun Song Shin. Measuring the incidence of insider trading in a market for state-contingent claims. *The Economic Journal*, 103(420):1141–1153, 1993. doi: 10.2307/2234240.

Martin Spann and Bernd Skiera. Sports forecasting: A comparison of the forecast accuracy of prediction markets, betting odds and tipsters. *Journal of Forecasting*, 28(1):55–72, 2009. doi: 10.1002/for.1091.

SportsPredict. Probability Cup: Scoring. https://sportspredict.com/probabilitycup/scoring, 2026.

StatsBomb. StatsBomb open data. https://github.com/statsbomb/open-data. World Cups 2018 and 2022, Euro 2020 and 2024, Copa América 2024, AFCON 2023.

Erik Štrumbelj. On determining probability forecasts from betting odds. *International Journal of Forecasting*, 30(4):934–943, 2014. doi: 10.1016/j.ijforecast.2014.02.008.

Philip E. Tetlock and Dan Gardner. *Superforecasting: The Art and Science of Prediction*. Crown, 2015.

Conor Walsh and Alok Joshi. Machine learning for sports betting: Should model selection be based on accuracy or calibration? *Machine Learning with Applications*, 16:100539, 2024. doi: 10.1016/j.mlwa.2024.100539.

Justin Wolfers and Eric Zitzewitz. Prediction markets. *Journal of Economic Perspectives*, 18(2):107–126, 2004. doi: 10.1257/0895330041371321.

John Yang, Carlos E. Jimenez, Alexander Wettig, Kilian Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir Press. SWE-agent: Agent-computer interfaces enable automated software engineering. In *Advances in Neural Information Processing Systems 37*, 2024. arXiv:2405.15793.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023. arXiv:2210.03629.

A Supplementary tables

Table 9. Walk-forward club backtest by question family: Brier score of the pricer, of a constant 50% and of the historical base rate, and the improvement over the base rate in Brier points (hundredths). Each family has about 41,163 forecasts.

Family	Pricer	50%	Base rate	Gain
Team has more shots on target	0.2136	0.2500	0.2493	+3.57
Team wins	0.2164	0.2500	0.2459	+2.94
Team has more corners	0.2268	0.2500	0.2488	+2.20
Team scores 2+	0.2236	0.2500	0.2454	+2.18
Team has 4+ shots on target	0.2097	0.2500	0.2301	+2.04
Team scores more in 2nd half	0.2171	0.2500	0.2331	+1.60
Team leads at halftime	0.2094	0.2500	0.2234	+1.40
Team has 5+ corners	0.2319	0.2500	0.2454	+1.35
Team scores in 2nd half	0.2361	0.2500	0.2470	+1.10
8+ total shots on target	0.2310	0.2500	0.2414	+1.04
3+ total goals	0.2416	0.2500	0.2503	+0.86
2+ goals in 2nd half	0.2418	0.2500	0.2471	+0.53
Both score and 3+ goals	0.2358	0.2500	0.2409	+0.51
Tied at halftime	0.2400	0.2500	0.2422	+0.22
10+ total corners	0.2482	0.2500	0.2503	+0.20
Team has more cards	0.2193	0.2500	0.2201	+0.09
More goals in 2nd half than 1st	0.2460	0.2500	0.2462	+0.02
Team commits more fouls	0.2459	0.2500	0.2459	−0.01
4+ total cards	0.2410	0.2500	0.2394	−0.15

Table 10. The 65 question families and how many of the 1,153 archived questions each covers.

Family	n	Family	n	Family	n
team_more_count_than	163	first_goal_and_other_scores_2h	9	header_goal	1
team_count_ge	159	team_first_goal_game	9	cards_gt_goals	1
player_sot_ge	113	team_scores_both_halves	6	plays_full_match	1
total_count_ge	94	team_first_goal_2h	5	gk_saves_ge	1
team_win	69	substitute_scores	5	total_subs_ge	1
total_goals	58	any_player_braces	5	team_any_player_sot_ge	1
player_scores_or_assists	51	shootout_decided	3	first_goal_not_players	1
player_scores	50	hold_lead	3	total_goals_odd	1
penalty_or_red	41	extra_time	3	first_goal_in_2h	1
team_scores_in_half	36	match_drawn	2	decided_by_one_goal	1
timing_event	34	red_card	2	either_wins_both_halves	1
btts_and_total_ge	29	team_win_by_ge	2	substitution_at_ht	1
ht_tied	26	clean_sheet	2	ht_scoreless	1
both_teams_count_ge	22	total_goals_exact	2	first_goal_single_digit	1
team_goals_ge	20	goal_each_half	2	team_first_substitution	1
btts	18	first_card_before_goal	2	var_review	1
h2_more_goals_than_h1	17	same_goals_halves	2	team_distinct_shooters_ge	1
advance	16	card_each_half	2	first_goal_assisted	1
ht_team_winning	14	any_player_sot_ge	1	penalty_scored	1
penalty_awarded	12	own_goal	1	player_more_sot_than	1
team_more_goals_2h	10	outside_box_goal	1	team_n_in_one_half	1
total_shots_ge	10	substitution_before_ht	1		